

Prelim 1 Review

HW5 Due Friday
6pm

Single Particles

- Particle Dynamics
- Zero-length spring
- Vector and Dyads
- later ← • Rotation Matrices

EOM {

- Conservation Lin/Ang. Momentum.
- Energy & Work
- Equilibrium

Matlab

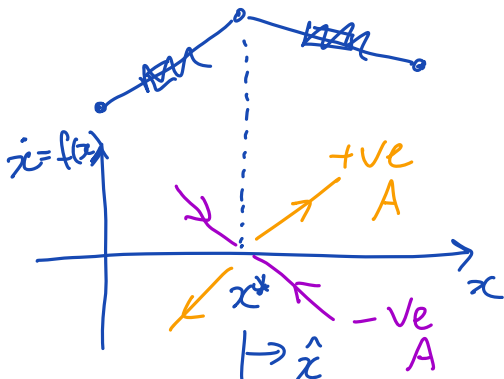
- Animating → Homework/Project.
- Ode45 } ✓
- fsolve.

Multiparticle/Rigid Body

- System Properties
- Inertia Tensor No 3D

Example 2 Jacobian

Stability occurs for -ve eigenvalues of A



$$\hat{x} = x - x^*$$

$$\dot{\hat{x}} = A\hat{x}$$

-ve, stable +ve, unstable

$$J_{ij} = \frac{\partial f_i}{\partial x_j} = \begin{bmatrix} \frac{\partial \dot{x}}{\partial x} & \frac{\partial \dot{x}}{\partial y} & \frac{\partial \dot{x}}{\partial v_x} & \frac{\partial \dot{x}}{\partial v_y} \\ \frac{\partial \dot{y}}{\partial x} & \frac{\partial \dot{y}}{\partial y} & \frac{\partial \dot{y}}{\partial v_x} & \frac{\partial \dot{y}}{\partial v_y} \\ \frac{\partial \dot{v}_x}{\partial x} & \frac{\partial \dot{v}_x}{\partial y} & \frac{\partial \dot{v}_x}{\partial v_x} & \frac{\partial \dot{v}_x}{\partial v_y} \\ \frac{\partial \dot{v}_y}{\partial x} & \frac{\partial \dot{v}_y}{\partial y} & \frac{\partial \dot{v}_y}{\partial v_x} & \frac{\partial \dot{v}_y}{\partial v_y} \end{bmatrix}$$

$$\frac{\partial \dot{x}}{\partial x} = \frac{\partial}{\partial x} (v_x) = 0$$

$$\frac{\partial \dot{x}}{\partial v_x} = \frac{\partial}{\partial v_x} (v_x) = 1$$

$$\frac{\partial \dot{v}_x}{\partial x} = \frac{\partial}{\partial x} \left(\frac{-k_A}{m} (x - r_{Ax} - L_{O,Ax}) \frac{(x - r_{Ax})}{\sqrt{(x - r_{Ax})^2 + (y - r_{Ay})^2}} \right) + \dots$$

@ $x \approx -5$
 $y = 0.174$

$$r_{Ax} = 3$$

$$L_{OAx} \approx 5$$

$$k_A = 50$$

$$r_{Bx} = -3$$

$$L_{OBx} \approx 5$$

$$k_B = 50$$

$$m = 0.1$$

$$= \frac{\partial}{\partial x} \left(\frac{-50}{0.1} (x - 8) - \frac{50}{0.1} (x - 2) \right)$$

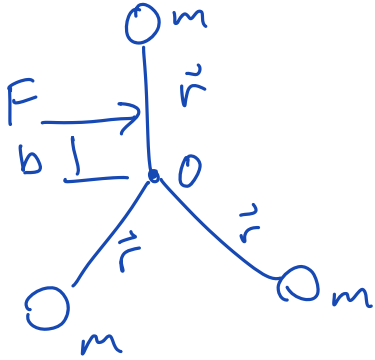
$$= \frac{-50}{0.1} [2]$$

$$= -1000$$

real value $-999.9 \dots$

Example 2 EoM

- Rigid body, 3 equal masses.
Fixed arms have negligible mass.

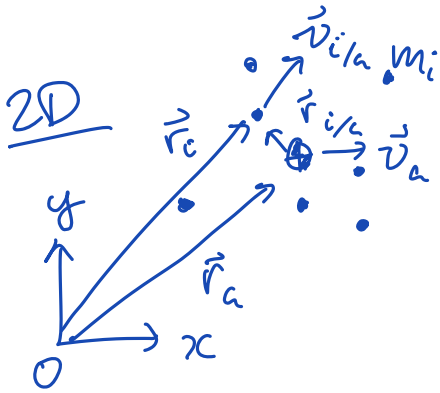


Force ' $\vec{F}$ ' applied at distance ' b ' from center.

- Find
- acceleration of O (a_o)
 - angular acceleration of frame. ($\vec{\omega}$)

Ans will be updated
here after weekend.

Example 3 - Energy



a) Find $E_k = E_{k/a} + E_{k/a}$

b) Simplify $E_{k/a}$ for rigid objects

$$a) E_k = \sum \frac{m_i}{2} \vec{v}_i^2$$

$$\vec{v}_i = \vec{v}_a + \vec{v}_{i/a}$$

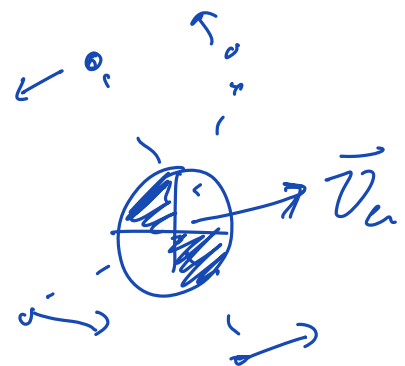
$$= \sum \frac{m_i}{2} (\vec{v}_a + \vec{v}_{i/a})^2$$

$$= \sum \frac{m_i}{2} (\vec{v}_a^2 + 2\vec{v}_a \vec{v}_{i/a} + \vec{v}_{i/a}^2)$$

$$= \underbrace{\sum \frac{m_i}{2} \vec{v}_a^2}_{E_{ka}} + \underbrace{\sum m_i \vec{v}_a \vec{v}_{i/a}}_{\text{zero.}} + \underbrace{\sum \frac{m_i}{2} \vec{v}_{i/a}^2}_{E_{k/a}}$$

$$\text{2nd term} = \vec{v}_a \sum m_i \vec{v}_{i/a}$$

$$\sum m_i (\vec{v}_i - \vec{v}_a)$$



see Lect 14
Demo.

